

WATERWATCH · REQUIREMENTS

# WaterWatch requirements

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DOCUMENT CONTROL

|             |                                         |         |                |
|-------------|-----------------------------------------|---------|----------------|
| Document ID | WWT-REQ-001                             | Version | 0.2            |
| Status      | Draft                                   | Date    | 2026-09-25     |
| Author      | Amish Chadha                            | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 2 Technology concept formulated     |         |                |
| Repository  | github.com/BoujeeEnjinia1701/waterwatch |         |                |
| Source      | docs/03-requirements.md @ 806e6e4       |         |                |

REVISION HISTORY

| Version | Date       | Author       | Change                                                       |
|---------|------------|--------------|--------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with concept status |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                             |

# WaterWatch requirements

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These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after co-design sessions. Two requirements are **not met** on present evidence (R1 and R10) and two are at risk (R2 and R9); see the status column and WWT-PRC-001.

Table 1. Requirements and concept status.

| ID  | Requirement                      | Target                                                                                                                                                                                                        | Verification (TRL 3 or later)                                                 | Concept status                                                                                                                              |
|-----|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Measure free chlorine at the tap | 0 to 2.0 mg/L; within $\pm 0.1$ mg/L below 1.0 mg/L and $\pm 15$ % above, at pH 6.5 to 8.5 and 5 to 40 °C                                                                                                     | Literature and sensor data at TRL 3; later side-by-side with a DPD comparator | <b>Not met</b> on evidence: the low-cost membrane-free sensor is unproven over months, and the reading depends on pH, which is not measured |
| R2  | Measure turbidity                | 0 to 100 NTU, reporting "above 100 NTU" beyond; within $\pm 0.5$ NTU or $\pm 10$ %, whichever is greater                                                                                                      | Comparison with the published open-source design; later formazin standards    | At risk: bench accuracy is published, but fouling and bubbles in continuous use are not                                                     |
| R3  | Measure water temperature        | 0 to 50 °C, within $\pm 0.5$ °C                                                                                                                                                                               | Sensor datasheet                                                              | Met by design                                                                                                                               |
| R4  | Sample on a schedule             | One reading per hour by default; configurable from 15 min to 24 h                                                                                                                                             | Firmware sketch review                                                        | Met by design                                                                                                                               |
| R5  | Alert the people who act         | SMS to up to three numbers within 15 min of a confirmed crossing (two consecutive readings): free chlorine below 0.2 mg/L, turbidity above 5 NTU, device fault or low battery; thresholds adjustable per site | Design review; later timed trial                                              | Met by design where there is cellular coverage                                                                                              |
| R6  | Report and keep data             | Upload readings at least every 4 h over LTE-M, NB-IoT or 2G (LoRaWAN variant); keep 90 days on board if the link fails; open CSV or JSON format; works without a cloud service                                | Data budget calculation                                                       | Met by design                                                                                                                               |
| R7  | Run on sunlight alone            | 7 days or more with no sun from full; back to full in 3 clear days or fewer                                                                                                                                   | Energy budget calculation                                                     | Met by estimate (about 30 days; about 1.2 days to recharge)                                                                                 |
| R8  | Use little treated water         | 20 L/day or less at hourly sampling                                                                                                                                                                           | Flow calculation                                                              | Met by estimate (about 18 L/day), thin margin                                                                                               |
| R9  | Survive outdoors                 | Electronics IP65 or better; ambient 0 to 45 °C in direct sun; UV-stable parts; flow cell opaque to daylight; battery charging blocked below 0 °C and above 45 °C                                              | Design review; later thermal estimate                                         | At risk: enclosure temperature in full sun not yet estimated                                                                                |
| R10 | Be maintainable by a caretaker   | One visit per month or less, 30 min or less, no special tools: clean the cell, compare with a DPD kit, adjust calibration from a phone or keypad                                                              | Maintenance task analysis; later field trial                                  | <b>Not met</b> on evidence: the chlorine sensor's drift and recalibration interval are unknown                                              |
| R11 | Install simply and safely        | Two people, 2 h or less, hand tools only; one tee and isolation valve on the riser; backflow prevented; wetted parts food-safe                                                                                | Design review                                                                 | Met by design                                                                                                                               |
| R12 |                                  |                                                                                                                                                                                                               | Priced BOM                                                                    | Met by estimate (about \$270)                                                                                                               |

| ID  | Requirement                       | Target                                                                               | Verification (TRL 3 or later)  | Concept status |
|-----|-----------------------------------|--------------------------------------------------------------------------------------|--------------------------------|----------------|
|     | Stay within the concept budget    | Parts \$300 or less per unit, excluding the shared calibration kit and airtime       |                                |                |
| R13 | Report measurements, not verdicts | Messages state what was measured and when; the device never says the water is "safe" | Design review of alert wording | Met by design  |

## ■ Assumptions

- Site: public tapstand on a piped, chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi) at the riser.
- Alert thresholds follow the field guidance summarized in WWT-PRB-001: at least 0.2 mg/L free chlorine at the point of delivery, and chlorination losing effectiveness as turbidity rises. The 5 NTU default is a conservative alert level, proposed, awaiting Amish and partner input.
- Free chlorine accuracy targets follow the resolution of common DPD comparator kits (about 0.1 mg/L), because the caretaker's DPD test is the field reference.
- pH affects the split between hypochlorous acid and hypochlorite, and amperometric sensors respond mainly to hypochlorous acid, so R1 assumes a site pH entered at installation and checked monthly unless a pH probe is added.
- Budget: the \$300 in `project.yaml` covers one unit's parts. A shared calibration kit (about \$60) and airtime (about \$1 to \$3 per month) are listed separately; both are estimates.